ITA0608 - Machine Learning
List of Lab Exercises

9. Compare Linear and Polynomial Regression using Python.

Aim

To compare the performance of **Linear Regression** and **Polynomial Regression** using Python.

Algorithm

1. Import the required Python libraries.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Train the Linear Regression model.
5. Transform the data into polynomial features.
6. Train the Polynomial Regression model.
7. Predict the test data using both models.
8. Compare their R^2 scores and display the results.

Dataset 1:

Program Code

```
from sklearn.datasets import load_diabetes

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.preprocessing import PolynomialFeatures


data = load_diabetes()

X = data.data

y = data.target
```

```python
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

linear_model = LinearRegression()
linear_model.fit(X_train, y_train)
linear_score = linear_model.score(X_test, y_test)

poly = PolynomialFeatures(degree=2)
X_train_poly = poly.fit_transform(X_train)
X_test_poly = poly.transform(X_test)

poly_model = LinearRegression()
poly_model.fit(X_train_poly, y_train)
poly_score = poly_model.score(X_test_poly, y_test)

print("Linear Regression R2 Score:", linear_score)
print("Polynomial Regression R2 Score:", poly_score)
```

Output:

```
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures

data = load_diabetes()
X = data.data
y = data.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

linear_model = LinearRegression()
linear_model.fit(X_train, y_train)
linear_score = linear_model.score(X_test, y_test)

poly = PolynomialFeatures(degree=2)
X_train_poly = poly.fit_transform(X_train)
X_test_poly = poly.transform(X_test)

poly_model = LinearRegression()
poly_model.fit(X_train_poly, y_train)
poly_score = poly_model.score(X_test_poly, y_test)

print("Linear Regression R2 Score:", linear_score)
print("Polynomial Regression R2 Score:", poly_score)
```

```
Linear Regression R2 Score: 0.4526027629719195
Polynomial Regression R2 Score: 0.4156399336408013
```

Dataset 2:

Program Code:

```
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
X = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10]).reshape(-1, 1)
y = np.array([18, 30, 43, 57, 68, 78, 86, 92, 96, 98])
linear_model = LinearRegression()
linear_model.fit(X, y)

new_sample = np.array([[7.5]])
linear_prediction = linear_model.predict(new_sample)
poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(X)
```

```
poly_model = LinearRegression()
poly_model.fit(X_poly, y)

new_sample_poly = poly.transform(new_sample)
poly_prediction = poly_model.predict(new_sample_poly)
print("DATASET 2: STUDENT MARKS PREDICTION")
print("-----------------------------------")
print("New Study Hours:", 7.5)
print("Linear Regression Prediction:", round(linear_prediction[0], 2), "%")
print("Polynomial Regression Prediction:", round(poly_prediction[0], 2), "%")

print("\nComparison:")
if poly_prediction[0] > linear_prediction[0]:
    print("Polynomial Regression predicts higher marks.")
else:
    print("Linear Regression predicts higher marks.")
```

Output:

```
else:
    print("Linear Regression predicts higher marks.")

... DATASET 2: STUDENT MARKS PREDICTION
-----------------------------------
New Study Hours: 7.5
Linear Regression Prediction: 85.07 %
Polynomial Regression Prediction: 88.29 %

Comparison:
Polynomial Regression predicts higher marks.
```

Dataset 3:

Program Code:

```
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
X = np.array([40, 50, 60, 70, 80, 90, 100, 110, 120, 130]).reshape(-1, 1)
y = np.array([4.5, 4.8, 5.2, 5.9, 6.8, 8.0, 9.5, 11.2, 13.4, 16.0])
linear_model = LinearRegression()
linear_model.fit(X, y)

new_sample = np.array([[95]])
linear_prediction = linear_model.predict(new_sample)
poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(X)

poly_model = LinearRegression()
poly_model.fit(X_poly, y)
```

```
new_sample_poly = poly.transform(new_sample)
poly_prediction = poly_model.predict(new_sample_poly)
print("DATASET 3: CAR FUEL CONSUMPTION")
print("--------------------------------")
print("New Speed:", 95, "km/h")
print("Linear Regression Prediction:",
      round(linear_prediction[0], 2), "L/100 km")
print("Polynomial Regression Prediction:",
      round(poly_prediction[0], 2), "L/100 km")

print("\nComparison:")
if poly_prediction[0] > linear_prediction[0]:
    print("Polynomial Regression predicts higher fuel consumption.")
else:
    print("Linear Regression predicts higher fuel consumption.")
```

Output:

```
        print("Polynomial Regression predicts higher fuel consumption.")
    else:
        print("Linear Regression predicts higher fuel consumption.")

... DATASET 3: CAR FUEL CONSUMPTION
    --------------------------------
    New Speed: 95 km/h
    Linear Regression Prediction: 9.78 L/100 km
    Polynomial Regression Prediction: 8.74 L/100 km

    Comparison:
    Linear Regression predicts higher fuel consumption.
```

Dataset 4:

Program Code:

```
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
X = np.array([5, 10, 15, 20, 25, 30, 35, 40, 45, 50]).reshape(-1, 1)
y = np.array([1.2, 2.5, 4.1, 6.2, 8.8, 11.9, 15.5, 19.6, 24.2, 29.3])
linear_model = LinearRegression()
linear_model.fit(X, y)

new_sample = np.array([[32]])
```

```
linear_prediction = linear_model.predict(new_sample)
poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(X)

poly_model = LinearRegression()
poly_model.fit(X_poly, y)

new_sample_poly = poly.transform(new_sample)
poly_prediction = poly_model.predict(new_sample_poly)
print("DATASET 4: ADVERTISING COST VS SALES")
print("------------------------------------")
print("New Advertising Cost: ₹", 32, "Thousand")
print("Linear Regression Prediction: ₹",
      round(linear_prediction[0], 2), "Lakhs")
print("Polynomial Regression Prediction: ₹",
      round(poly_prediction[0], 2), "Lakhs")

print("\nComparison:")
if poly_prediction[0] > linear_prediction[0]:
    print("Polynomial Regression predicts higher sales.")
else:
    print("Linear Regression predicts higher sales.")
```

Output:

```
    print("Polynomial Regression predicts higher sales.")
else:
    print("Linear Regression predicts higher sales.")

... DATASET 4: ADVERTISING COST VS SALES
    ------------------------------------
    New Advertising Cost: ₹ 32 Thousand
    Linear Regression Prediction: ₹ 15.13 Lakhs
    Polynomial Regression Prediction: ₹ 13.3 Lakhs

    Comparison:
    Linear Regression predicts higher sales.
```

Dataset 5:

Program Code:

```
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
X = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10]).reshape(-1, 1)
y = np.array([3.0, 3.8, 4.9, 6.3, 8.0, 10.2, 12.9, 16.1, 19.8, 24.0])
linear_model = LinearRegression()
linear_model.fit(X, y)

new_sample = np.array([[7.5]])
linear_prediction = linear_model.predict(new_sample)
poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(X)

poly_model = LinearRegression()
poly_model.fit(X_poly, y)

new_sample_poly = poly.transform(new_sample)
poly_prediction = poly_model.predict(new_sample_poly)
print("DATASET 5: EXPERIENCE VS SALARY")
print("--------------------------------")
print("New Experience:", 7.5, "years")
print("Linear Regression Prediction: ₹",
      round(linear_prediction[0], 2), "Lakhs")
print("Polynomial Regression Prediction: ₹",
      round(poly_prediction[0], 2), "Lakhs")

print("\nComparison:")
if poly_prediction[0] > linear_prediction[0]:
    print("Polynomial Regression predicts higher salary.")
else:
    print("Linear Regression predicts higher salary.")
```

Output:

```
    print("Polynomial Regression predicts higher salary.")
else:
    print("Linear Regression predicts higher salary.")
```

```
DATASET 5: EXPERIENCE VS SALARY
------------------------------
New Experience: 7.5 years
Linear Regression Prediction: ₹ 15.49 Lakhs
Polynomial Regression Prediction: ₹ 14.56 Lakhs

Comparison:
Linear Regression predicts higher salary.
```

Result:

The **Linear Regression** and **Polynomial Regression** models were successfully implemented and compared using Python. Their performances were evaluated using the **R^2 score**. For this dataset, **Linear Regression** achieved a slightly higher R^2 score than **Polynomial Regression**, indicating better performance.